

# EDUARDO DE JESÚS DÁVILA MEZA, PH.D.

AI/ML • Computer Vision • Embedded Systems • ROS/ROS2

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🎓 Federal Professional Certificates: Bachelor's Degree: 12027207 [✉](#) • Master's Degree: 14043743 [✉](#) • Ph.D. Degree: 15067339 [✉](#)

## TECHNICAL SKILLS

**AI/ML:** Computer Vision, Predictive Modeling, LSTM, Time Series Analysis (ARIMA), Deep Learning (Mask R-CNN), Model Optimization  
**Programming Languages:** Python, C++, SQL, Bash, MATLAB  
**Frameworks & Libraries:**  
**C++ and Python:** OpenCV, TensorFlow/Keras  
**Python:** Scikit-learn, Pandas, Matplotlib, Seaborn, NumPy, JSON, Tkinter  
**Deployment & Backend:** FastAPI, SQLAlchemy  
**Robotics Frameworks:** ROS/ROS2 with C++ and Python, RViz, Gazebo  
**Tools:** Ubuntu (Linux), VS Code, Git, GitHub, GitLab, Jupyter Notebook  
**Document Preparation and Office Tools:** LaTeX, Markdown, MediaWiki, MS Office, Dia (technical diagram editor)

## WORK EXPERIENCE & | [FULL LIST](#) [✉](#)

### Programming Instructor – ROS/ROS2

April 2024 – Present

*Tecnológico de Monterrey (ITESM), Guadalajara*

*Zapopan, Jalisco*

1) Designed and delivered hands-on courses in ROS/ROS2 using C++ and Python, focused on differential drive robots. 2) Developed practical repositories and lecture notes covering the foundations of development, simulation, deployment, and monitoring of ROS/ROS2-based modular systems, including computer vision and autonomous navigation. 3) Recognized as a top-rated instructor (See recognition [✉](#)). See repository 2025 [✉](#). See latest repository [✉](#).

✚ C++ • Python • ROS/ROS2 • Bash • Computer Vision • Robotics • Technical instruction

### Mathematical Analyst in Information Technologies

September 2025 – April 2026

*Geovoy, Busmen Group*

*Tlajomulco de Zúñiga, Jalisco*

1) Built an automated reporting tool using LLM-powered APIs to generate analytical reports and insights. 2) Developed statistical and machine learning models (time series analysis (ARIMA), LSTM) for route optimization and demand prediction, generating insights to improve operational decision-making. 3) Designed and optimized APIs using FastAPI, raw SQL, and Object-Relational Mapping (ORM) to integrate statistical models into production systems for geolocation and inventory management. 4) Deployed a MediaWiki-based platform to standardize internal processes, code, and API documentation.

✚ Python • FastAPI • SQLAlchemy • Time Series Analysis • LSTM • Forecasting • Predictive Modeling • GPT-OSS-20B • Data Analysis • Bash

### AI/DL Research Engineer – Computer Vision

May 2021 – July 2024

*Cinvestav, Guadalajara: Medical Image Segmentation Project*

*Zapopan, Jalisco*

1) Developed a deep learning model for medical image analysis, in collaboration with German research institutions, to support clinical decision-making, focusing on segmentation and quantification of retinal pathologies. 2) Applied data preprocessing, augmentation, feature extraction, and model validation techniques to ensure robustness and reliability in healthcare-related datasets. 3) Worked with high-resolution image datasets to improve the detection and segmentation of retinal pathologies. 4) Contributed to a research publication in computer vision and medical imaging.

✚ Computer Vision • Deep Learning • TensorFlow/Keras • Mask R-CNN • Image Segmentation • Data labeling, augmentation, and visualization

### Computer Vision & ROS Developer

December 2019 – October 2023

*Cinvestav, Guadalajara: Intelligent Visual Guide System (OJO SMART) — Modular Navigation Device*

*Zapopan, Jalisco*

1) Developed ROS-based modules for real-time recognition of colors, objects, signs, banknotes, and text using computer vision techniques. 2) Contributed to the integration of multiple perception systems into an assistive navigation device for visually impaired users.

✚ Computer Vision • C++ • Python • ROS • Tesseract OCR • OpenCV • TensorFlow

## PUBLICATIONS & PATENTS

**Meeting Abstract | June 2024** | “Deep-learning based quantification of RPE65-mutation inherited retinal degeneration”, presented at *Investigative Ophthalmology & Visual Science*, vol. 65(7), 1392, [DOI](#) ID: 2794864 [✉](#).

✚ AI/ML lifecycle • Computer vision / Medical image analysis • Data visualization • Mask R-CNN • Feature extraction • Research

**Journal Article | September 2023** | “Quaternion and Split Quaternion Neural Networks for Low-Light Color Image Enhancement”, in *IEEE Access*, vol. 11, 108257-108280, [DOI](#) 10.1109/ACCESS.2023.3312234 [✉](#).

✚ AI/ANN lifecycle • Computer vision / Image color analysis • Quaternion algebras • Color spaces • EKF

**Patent | March 2017** | “Device for controlling underactuated two-link systems with one actuator”, filed under the Invention Support Program, University of Guadalajara. Application no. MX/a/2017/016436.

✚ Embedded systems • Control theory • Digital and power electronics • PICs • SPI & I2C communication protocols

## ACADEMIC DEGREES

### Ph.D. in Science — AI/Deep Learning | Cinvestav, Guadalajara

September 2019 – May 2024

**Thesis** | Deep learning for recognition and quantification of fundus pathologies using instance segmentation, and quaternion neural networks for low-light image enhancement. Download thesis [✉](#).

✚ AI/ML lifecycle • Computer vision / Image analysis • Deep Learning • CNN/NN • Robotics • Research • Science Communication

### M.Sc. — AI/Machine Learning | Cinvestav, Guadalajara

September 2017 – August 2019

**Thesis** | Quaternion neural networks for low-light image enhancement, and identification of an electromechanical system. Download thesis [✉](#).

✚ AI/ANNs • Computer vision • Linear algebra • Control theory • Robotics • Probability & statistics • Research • Science Communication

### B.Eng. in Mechatronics — Embedded Systems | University of Guadalajara

August 2012 – December 2016

**Social Service & Professional Internship** | Developed embedded and mechatronic projects in an electronics and telecommunications laboratory.

✚ Embedded systems • Linear algebra • Calculus • Control theory • Digital and power electronics • HMI • PICs & Arduino

## CERTIFICATIONS | [ADDITIONAL TRAINING](#)

AI/DL [✉](#)  
Python [✉](#)

AI/ML [✉](#)  
R for DS [✉](#)

Data Cleaning [✉](#)  
ROS [✉](#)

LLMs [✉](#)  
Scikit-learn [✉](#)

NLP [✉](#)  
SQL [✉](#)

OpenCV [✉](#)  
TF-Keras [✉](#)

Pandas [✉](#)  
Full list [✉](#)